Avanish Mishra, Ph.D.

Personal Data

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Current Positions

Nov 2023-Present	Staff Scientist 2, **Theoretical Division (T-1), Physics and Chemistry of Materials, Los Alamos National Laboratory**, Los Alamos, NM, USA

Professional Summary

Computational materials scientist specializing in the design of structural materials for extreme environments and functional materials for energy-to-electronics applications. My work integrates first-principles methods and molecular dynamics with emerging paradigms such as machine learning and quantum computing to accelerate materials discovery. I have contributed to open data infrastructure (e.g., aNANt database, JARVIS-Leaderboard, URSA) and authored over 38 peer-reviewed publications.

Expertise

- **Scientific Expertise:** Computational materials science, quantum materials, structural and functional materials, high-throughput screening, materials informatics, and quantum chemistry.
- **Modeling and Simulation Tools:** Density functional theory (DFT), molecular dynamics (MD), and related packages such as VASP, LAMMPS, Quantum ESPRESSO, pySCF, Qiskit, and ASE; workflow automation for large-scale simulations, database generation, and the development of machine-learned interatomic potentials (MLIAP).
- **Machine Learning and AI:** Materials-focused machine learning (ML), generative AI (GenAI), large language models (LLMs), Agentic AI, active learning for accelerated materials discovery.

Education

Aug 2014-Feb 2019	**PhD in Materials Science**, **Materials Research Centre, Indian Institute of Science**, Bengaluru, India **Dissertation:** Exploration of exfoliation, functionalization, and properties of MXenes via first-principles and machine learning
July 2013	**Master of Science** in Physics **Deen Dayal Upadhyaya Gorakhpur University**, Gorakhpur, India
July 2011	**Bachelor of Science** in Physics, Math, and Chemistry **Deen Dayal Upadhyaya Gorakhpur University**, Gorakhpur, India

Research experience

Staff Scientist 2 Nov 2023-Present	Physics and Chemistry of Materials (T-1), Theoretical Division Los Alamos National Laboratory, Los Alamos, NM, USA
Director's Postdoc Oct 2022-Nov 2023	Los Alamos National Laboratory, Los Alamos, NM, USA Sponsors: Dr. Edward M. Kober and Dr. Nithin Mathew
Postdoctoral Fellow Jan 2022-Sep 2022	Los Alamos National Laboratory, Los Alamos, NM, USA Mentors: Dr. Edward M. Kober and Dr. Nithin Mathew
Research Scientist May 2021-Dec 2021	University of Connecticut, Storrs, CT, USA Mentor: Prof. Avinash M. Dongare
Postdoctoral Fellow May 2019-April 2021	University of Connecticut, Storrs, CT, USA Mentor: Prof. Avinash M. Dongare
Graduate Researcher Aug 2014-Feb 2019	Indian Institute of Science, Bengaluru, KA, India Advisor: Prof. Abhishek K. Singh

Scholarships and Awards

1. One of the 30 postdocs within the Lab selected for "Science in 3" 2023 at Los Alamos National Laboratory (LANL), **June 2023**
2. **Fellowship** from Los Alamos National Laboratory (LANL), USA, to carry out research as Director's postdoctoral fellow at LANL, USA, **Aug 2022**
3. **Travel award** from Department of Science and Technology, India to attend American Physical Society March Meeting, USA, **Feb 2019**
4. **Travel award** from Group on Energy Research and Applications (GERA) to attend American Physical Society March Meeting, USA, **Feb 2019**
5. **Travel award** to attend MATERIALS 4.0 workshop at Technical University of Dresden, Dresden, Germany, **Sep 2018**
6. **Travel award** from Group on Energy Research and Applications (GERA) to attend American Physical Society March Meeting, USA, **Feb 2018**
7. **Travel award** to attend advanced workshop at International Centre for Theoretical Physics, Trieste, Italy, **Jan 2017**
8. **Kawazoe Prize, Best Poster Award** in ACCMS-TM 2016 at SRM University Chennai, India, **Sep 2016**
9. **Qualified joint CSIR-UGC National Eligibility Test (NET)** 2013 (December) with all India rank of **58** in Physical Science.
10. **Qualified Graduate Aptitude Test in Engineering (GATE)** 2014 with all India rank of **36** in Physical Science.

Publications

Total **2538 citations and h-index of 19** (source: Google scholar at July 7, 2026)

Peer-reviewed articles:

1. **A. Mishra**, P. Srivastava, H. Mizuseki, K-R. Lee and Abhishek K. Singh, "*Isolation of pristine MXene from Nb_4AlC_3 MAX phase: A first-principles Study*" **Phys. Chem. Chem. Phys. 18, 11073 (2016)**
2. P. Srivastava*, **A. Mishra***, H. Mizuseki, K-R. Lee and Abhishek K. Singh, "*Mechanistic Insight into the Chemical Exfoliation and Functionalization of Ti_3C_2 MXene*", ACS Appl. Mater. Interfaces. 8, 24256 (2016) | *Equal contribution

3. **A. Mishra**, P. Srivastava, A. Carreras, I. Tanaka, H. Mizuseki, K-R. Lee and Abhishek K. Singh,, "*Atomistic Origin of Phase Stability in Oxygen Functionalized MXene: A Comparative Study*", **J. Phys. Chem. C 121, 18947 (2017)**

4. A. Chandrasekaran, **A. Mishra**, and Abhishek K. Singh, "*Ferroelectricity, Antiferroelectricity, and Ultrathin 2D Electron/Hole Gas in Multifunctional Monolayer MXene*", **Nano Lett., 17, 3290 (2017)**

5. R. Koizumi, S.Ozden, A. Samanta, A. P. P. Alves, **A. Mishra**, G. Ye, G. G. Silva, R. Vajtai, Abhishek K. Singh, C. S. Tiwary, and P M. Ajayan, "*Origami-Inspired 3D Interconnected Molybdenum Carbide Nanoflakes*" **Adv. Mater. Interfaces 5, 1701113 (2018)**

6. A. C. Rajan*, **A. Mishra***, S. Satsangi, R. Vaish, H. Mizuseki, K-R. Lee and Abhishek K. Singh, "*Machine-Learning-Assisted Accurate Band Gap Predictions of Functionalized MXene.* **Chem. Mater 30, 4031 (2018)** | *Equal contribution
-Mentioned as a seminal article in Virtual Issue on Machine-Learning Discoveries in Materials Science of Chemistry of Materials

7. **A. Mishra***, S. Satsangi*, A. C. Rajan, H. Mizuseki, K-R. Lee and Abhishek K. Singh, "*Accelerated Data-Driven Accurate Positioning of the Band-Edges of MXenes.*" **J. Phys. Chem. Lett. 10, 780 (2019)** | *Equal contribution

8. M. Khazaei, **A. Mishra**, N. S. Venkataramanan, Abhishek K. Singh and S. Yunok, "*Recent Advances in MXenes: From Fundamentals to Applications.*" **Curr. Opin. Solid State Mater. Sci. 23, 164 (2019)**

9. N. Sethulakshmi, **A. Mishra**, P. M. Ajayan, Y. Kawazoe, A. Roy, Abhishek K. Singh and C. S. Tiwary, "*Magnetism in Two-Dimensional Materials Beyond Graphene.*" **Mater. Today 27, 107 (2019)**

10. N. Lertcumfu, Farheen N. Sayed, Sharmila N. Shirodkar, S. Radhakrishnana, **A. Mishra**, G. Rujijanagul, Abhishek K. Singh, Boris. I. Yakobson, C. S. Tiwary, and P. M. Ajayan "*Structure-Dependent Electrical and Magnetic Properties of Iron Oxide Composites.*" **Physica Status Solidi (A) 1801004, (2019)**

11. T. Pandey, A. Nissimagoudar, **A. Mishra** and Abhishek K. Singh, "*Ultralow Thermal Conductivity and Anomalously High Thermoelectric Figure of Merit in Ternary In_5X_5Br (X = S, and Se) Compounds.*" **J. Mater. Chem. A 8, 13812 (2020)**

12. S. Parida, **A. Mishra**, J. Chen, J. Wang, C. B. Carter, and Avinash M. Dongare, "*Vertically Stacked 2H-1T Dual-phase MoS_2 Microstructures During Lithium Intercalation: A first Principles Study*" **J. Am. Ceram. Soc 103, 6603 (2020)**

13. S. N. Shuvo, A. M. U. Gomez, **A. Mishra**, W. Y. Chen, Avinash M. Dongare, and L. A Stanciu, "*Sulfur-doped Titanium Carbide MXenes for Room Temperature Gas Sensing*" **ACS Sens. 5, 2915 (2020)**

14. P. Xu, W. Rheinheimer, **A. Mishra**, S. N. Shuvo, Z. Qi, H. Wang, Avinash M. Dongare, and L. A Stanciu, "*Origin of High Interfacial Resistance in Solid-State Batteries: LLTO/LCO Half-cells*" **ChemElectroChem 8, 1847 (2021)**

15. **A. Mishra**, C. Kunka, M. J. Echeverria, R. Dingreville, and Avinash M. Dongare, "*Fingerprinting Shock-induced Deformations via Diffraction.*" **Sci. Rep. 11, 9872 (2021)**

16. M. J. Echeverria, S. Galitskiy, **A. Mishra**, and Avinash M. Dongare, "*Modeling the Evolution of Microstructure during Laser Shock Loading and Spall Failure of Cu Microstructures at the Atomic Scales* " **Comput. Mater. Sci. 198, 110668 (2021)**

17. **A. Mishra**, J. Lind, M. Kumar, and Avinash M. Dongare, "*Understanding the Phase Transformation Mechanisms that affect the Dynamic Response of Fe-based Microstructures at the Atomic Scales*" **JAP 130, 215902 (2021)**

18. S. Satsangi*, **A. Mishra***, and Abhishek K. Singh, "*Feature Blending: An Approach Towards Unified Machine Learning Models for Band Gap Prediction.*" **ACS Phys. Chem. Au 2, 16 (2022)** | *Equal contribution

19. **A. Mishra**, M. J. Echeverria, K. Ma, S. Parida, C. Chen, S. Galitskiy, and Avinash M. Dongare, "*Virtual Texture Analysis to Investigate the Deformation Mechanisms in Metal Microstructures at the Atomic-Scale*" **J. Mater. Sci. 57, 10549 (2022)**

    -Graphical abstract selected as cover art

20. **A. Mishra**, K. Ma, and Avinash M. Dongare, "*Virtual Diffraction Simulations using the Quasi-Coarse-Grained Dynamics Method to Understand and Interpret Plasticity Contributions during In Situ Shock Experiments*" **J. Mater. Sci. 57, 12782 (2022)**

21. C. Ching, S. Galitskiy, **A. Mishra**, and Avinash M. Dongare, "*Modeling Laser Interactions with Aluminum and Tantalum Targets using a Hybrid Atomistic-Continuum Model*" **JAP 133, 105901 (2023)**

22. S. Galitskiy, **A. Mishra**, and Avinash M. Dongare, "*Modeling Shock-induced Void Collapse in Single-crystal Ta Systems at the Mesoscales*" **Int. J. Plast. 164, 103596 (2023)**

23. R. Kumar, J. Chen, **A. Mishra**, and Avinash M. Dongare, "*Interface Microstructure Effects on Dynamic Failure Behavior of Layered Cu/Ta Microstructures*" **Sci. Rep 13, 11365 (2023)**

24. **A. Mishra**, K. Dang, E. M. Kober, S. J. Fensin, and N. Mathew, "*Role of Microscopic Degrees of Freedom on Nanopillar Compression of Bicrystal Cu*" **Mater. Res. Lett. 11, 872 (2023) | (LA-UR-23-20212)**

25. S. Parida, **A. Mishra**, Q. Yang, A. Dobley, C. B. Carter, and Avinash M. Dongare, "*Data-driven search for promising intercalating ions and layered materials for metal-ion batteries.*" **J. Mater. Sci. 59, 932 (2024)**

26. K. Choudhary et al., "*Large Scale Benchmark of Materials Design Methods*" **npj Comp. Mater. 10, 93, (2024)**

27. K. Dang, **A. Mishra**, S Suresh, E. M. Kober, N. Mathew, and S. J. Fensin "*Interplay between dislocation type and local structure in dislocation-twin boundary reactions in Cu*" **Phys. Rev. Mater. 8, 063604, (2024)**

28. T. M. Kucinski, R. Dhall, B. Savitzky, C. L. Ophus, R. Karkee, **A. Mishra**, E. Dervishi, J. H. Kang, C. Lee, J. Yoo, and M. T. Pettes "*Direct measurement of the thermal expansion coefficient of epitaxial WSe_2 by four-dimensional scanning transmission electron microscopy*" **ACS Nano 18, 27, (2024)**

29. A. Bärtschi, F. Caravelli, C. Coffrin, J. Colina, S. Eidenbenz, A. Jayakumar, S. Lawrence, M. Lee, A. Y. Lokhov, **A. Mishra**, S. Misra, Z. Morrell, Z. Mughal, D. Neill, A. Piryatinski, A. Scheie, M. Vuffray, Y. Zhang "*Potential Applications of Quantum Computing at Los Alamos National Laboratory*" **arXiv preprint** | *names are alphabetical

30. I. Chesser, P. M. Derlet, **A. Mishra** , S. Paguaga, N. Mathew, K. Q. Dang, B. P. Uberuaga, A. Hunter, S. Fensin "*The structure and migration of heavily irradiated grain boundaries and dislocations in Ni in the athermal limit*" **Phys. Rev. Mater. 8,093606 (2024)**

31. **A. Mishra**, S. Suresh, S. J. Fensin, E. M. Kober, and N. Mathew, "*Learning from metastable symmetric-tilt grain boundaries using physics-based descriptors*" **Phys. Rev. Mater. 8, 123605 (2024)**

32. R. Kumar, A. K. Shringi, H. J. Wood, S. Oturak, T. S. K. Sharma, R. Chaurasiya, **A. Mishra**, W. M. Choi, N. Y. Doumon, I. Daboc, M. Terrones, F. Yan, "*Substitutional doping of 2D transition metal dichalcogenides for device applications: Current status, challenges and prospects*" **Mater. Sci. Eng. R Rep: R 163, 100946 (2025)**

33. K. Dang, S. Suresh, **A. Mishra**, I. Chesser, N. Mathew, E. M. Kober, and S. J. Fensin, "*Dislocation-Grain Boundary Interaction Dataset for FCC Cu*" **Sci. Data 12, 955 (2025)**

34. M S. Nitol, **A. Mishra**, S. Xu, and S. J. Fensin, "*Moment tensor potential and its application in the Ti-Al-V multi-component system*" **Phys. Rev. Mater. 9, 063601 (2025)**

35. **A. Mishra**, Brenden W Hamilton, Mashroor S Nitol, Nithin Mathew, Timothy C. Germann, and S. J. Fensin, "*Unveiling Process–Structure Mapping with Generative AI*" **Proceedings of the IEEE/CVF International Conference on Computer Vision, 3593-3602 (2025)**

36. P. Tsurkan, M. J. Echeverria, **A. Mishra**, and Avinash M. Dongare, "*Influence of nanoscale interfaces on the dynamic deformation and spall failure of Cu–Fe alloy microstructures*" **JAP 138, 225902 (2025)**

37. B. Murgas, **A. Mishra**, N. Mathew, and A. Hunter, "*Phase field dislocation dynamics study of grain boundary-dislocation interactions.*" **JAP 139, 155101 (2026)**

38. **A. Mishra**, J. S. Carpenter, and S. J. Fensin, "*Controlling Deformation in Al/Ti: How Interface Roughness and Rotation Drive Bimetal Mechanics*" **Phys. Rev. Mater. 10, 063602 (2026)**

Upcoming journal articles:

1. Janith Wanni, Bernard Gaskey, Mikayla Obrist, Benjamin Lewis Neuman, Ayden Micheal Caulder, **A. Mishra***, Nithin Mathew*, Timothy C. Germann, Saryu J. Fensin, "*Data-Driven Discovery, Processing Optimization, and Experimental Validation of Novel High-Entropy Alloys*" **(Under preparation)**

2. Mashroor S. Nitol, **A Mishra**, Christopher D. Barrett, and Saryu Fensin, "*Active Learning Moment Tensor Potential for Solute and Defect Energetics in MgAlZn Alloys*" **(Under preparation)**

3. **A Mishra**, Nithin Mathew, Timothy C. Germann, and Saryu Fensin, "*VISTA: Vision Transformer for Structure-property Translation, Micrographs to Property*" **(Under preparation)**

4. Rahul Somasundaram, Adela Habib, Khanh Dang, Sachin Shivakumar, Ryley G. Hill, Golo Wimmer, **A Mishra**, Aleksandra Pachalieva, Arthur Lui, Hari Viswanathan, Michael Grosskopf, Saryu Fensin, Russell Bent, Nathan DeBardeleben, and Earl Lawrence, "*An Agentic Orchestration of Atomistic Simulations*" **(Under preparation)**

5. Ryley G. Hill, Sachin Shivakumar, Golo A. Wimmer, Rahul Somasundaram, Adela Habib, Nithin Mathew, **A Mishra**, Mashroor Nitol, Aleksandra Pachalieva, Hari Viswanathan, Dan O'Malley, Russell Bent, Michael Grosskopf, Abigail Hunter, Earl Lawrence, "*Autonomous Discovery of Materials Failure by Agentic Workflows*" **(Under preparation)**

6. **A. Mishra**, J. S. Carpenter, and S. J. Fensin, "*Effect of Interface Roughness and Orientation on Shock Deformation in Bimetallic Microstructures*" **(Under preparation)**

7. Matthew Maron, **A. Mishra**, Nithin Mathew, Giacomo Po, "*Data-Driven Upscaling of a Local Mobility Law: Atomistics to Dislocation Dynamics*" **(Under preparation)**

Teaching experience

Teaching assistant Aug-Dec 2015	MR 301: Quantum Mechanical Principles in Materials. Course instructor: Prof. Abhishek Kumar Singh Indian Institute of Science
Teaching assistant Jan-May 2016	MR 308: Computational modeling of materials. Course instructor: Prof. Abhishek Kumar Singh Indian Institute of Science
Co-Instructor Sep-Dec 2021	MSE 6401: Graduate Seminars Materials Science and Engineering University of Connecticut

Service and Leadership

Symposium Organizer* March 2027	TMS 2027 Annual Meeting & Exhibition 2027 Co-organizer of a symposium titled Artificial Intelligence Applications in Integrated Computational Materials Engineering (AI-ICME) | *upcoming

Symposium Organizer | TMS 2026 ANNUAL MEETING & EXHIBITION 2026
MARCH 2026 | Co-organizer of a symposium titled Artificial Intelligence Applications in Integrated Computational Materials Engineering (AI-ICME)

Symposium Organizer | TMS 2025 ANNUAL MEETING & EXHIBITION 2025
MARCH 2025 | Co-organizer of a symposium titled Artificial Intelligence Applications in Integrated Computational Materials Engineering

Symposium Organizer | MACH CONFERENCE 2022
APRIL 2022 | Co-organizer of three-day symposium titled Real-time characterization of materials under dynamic deformation

Theoretical Division | P/T (PHYSICS/THEORY) COLLOQUIUM COMMITTEE
MAY 2022 -Present | Member from Theoretical Division for P/T Colloquium LANL

PROFESSIONAL AFFILIATIONS

- Member, The Minerals, Metals & Materials Society (TMS)
- Member, Integrated Computational Materials Engineering (ICME)
- Member, American Physical Society (APS)
- Member, Materials Research Society (MRS)

REVIEWER OF JOURNALS

- Physical Review Letters, Physical Review B, Physical Review Applied, Physical Review Materials, Journal of Physical Chemistry Letters, npj Computational Materials, Scientific Reports, Scientific Data, Advanced Functional Materials, International Journal of Plasticity, Journal of Applied Physics, Journal of Materials Science, Sensors, Modelling and Simulation in Materials Science and Engineering, and Materials

PRESENTATIONS

Invited presentation

MAY 2026 *"Generative AI–Enabled Process–Structure–Property Optimization for Accelerated Materials Design and Discovery"* **A. Mishra**, *"Machine Learning in Chemical and Materials Sciences 2026"*, Santa Fe, NM, USA.

NOV 2025 *"Accelerating Materials Discovery via Generative AI"* **A. Mishra**, *"Los Alamos - Arizona Days Fall 2025"*, Los Alamos, NM, USA..

OCT 2025 *"Accelerating Materials Discovery via Generative AI"* **A. Mishra**, *"Applied AI Seminar Series, Argonne National Laboratory"*, Virtual.

SEP 2025 *"Accelerating Materials Discovery via Generative AI"* **A. Mishra**, *"Machine Learning and Informatics for Chemistry and Materials, Telluride Workshop"*, Telluride, Co, USA.

JULY 2025 *"Unveiling Process-Structure Mapping with Generative AI"* **A. Mishra**, *"Artificial Intelligence and Machine Learning for Materials"*, Santa Fe, NM, USA.

June 2025 “*Unveiling Process-Structure Mapping with Generative AI*” **A. Mishra**, *“Algorithms For Multiphysics Models In The Post-Moore’s Law Era,”*, Los Alamos, NM, USA.

May 2025 “*Bridging the Missing Link: Process–Structure Mapping in Materials Design with Generative AI*” **A. Mishra**, *“Machine Learning in Chemical and Materials Sciences 2025”*, Santa Fe, NM, USA.

May 2024 “*Learning from ‘Small’ Data Using Physics-Based Descriptors*” **A. Mishra**, *“Machine Learning in Chemical and Materials Sciences 2024”*, Virtual.

May 2024 “*Learning from ‘Small’ Data Using Physics-Based Descriptors*” **A. Mishra**, *“T-4 group’s seminar, BLABS, Theoretical Division, Los Alamos National Laboratory”*, Los Alamos, NM, USA.

March 2024 “*Insight into exfoliation, functionalization, and properties of MXenes via first-principles and machine learning*” **A. Mishra**, *Telluride workshop on “Quantum Computing for Quantum Chemistry, Molecular Dynamics, and Beyond”*, Telluride, CO, USA.

Oct 2023 “*Virtual Characterization Tools for Understanding Material Response in Harsh Conditions*” **A. Mishra**, *MS&T23 Technical Meeting and Exhibition*, Columbus, OH, USA.

Feb 2021 “*Understanding Plasticity Contributions from Slip, Twinning, and Phase Transformation under Shock Loading using Atomistic Simulations*” **A. Mishra**, *Department of Materials Science and Engineering*, University of Connecticut, Connecticut, USA.

Oral presentation

Mar 2026 “*Generative AI–Driven Process–Structure–Property Optimization for Materials Discovery*” **A. Mishra**, B. Hamilton, M. Nitol, A. Debnath; N. Mathew, T. Germann, and S. Fensin *2026 TMS Annual Meeting*, San Diego Convention Center in San Diego, California, USA.

June 2025 “*Effect of Interface Roughness and Orientation on Shock Deformation in Bimetallic Microstructures*” **A. Mishra**, J. Carpenter, and S. J. Fensin *24th Biennial Conference of the APS Topical Group on Compression of Condensed Matter* , Washington Hilton, Washington, DC, USA.

Dec 2024 “*Shaping Deformation—How Interface Roughness and Alignment Drive Bimetal Mechanics*” **A. Mishra**, J. Carpenter, and S. J. Fensin *2024 MRS Fall Meeting*, Hynes Convention Center, Boston, Massachusetts, USA.

Mar 2023 “*Predicting Grain Boundary Properties using Strain Functional Descriptors and Supervised Machine Learning*” **A. Mishra**, S. Suresh, K. Dang, S. J. Fensin, E. M. Kober, and N. Mathew, *2023 TMS Annual Meeting*, San Diego Convention Center in San Diego, California, USA.

Dec 2022 “*Structure-Property Relationships for Grain Boundaries Using Strain Functional Descriptors and Supervised Machine Learning*” **A. Mishra**, S. Suresh, K. Dang, S. J. Fensin, E. M. Kober, and N. Mathew, *2022 MRS Fall Meeting*, Hynes Convention Center, Boston, Massachusetts, USA.

Dec 2022 "*Analysis of Grain Boundary (GB) Structure and Dislocation-GB Interactions Using Unsupervised Machine Learning*" N. Mathew, **A. Mishra**, S. Suresh, K. Dang, S. J. Fensin, and E. M. Kober, *2022 MRS Fall Meeting*, Hynes Convention Center, Boston, Massachusetts, USA.

Oct 2022 "*Accurate Prediction of Grain Boundary Properties using Machine Learning and Strain Functional Descriptors*" **A. Mishra**, S. Suresh, K. Dang, S. J. Fensin, E. M. Kober, and N. Mathew, *2022 SES Annual Meeting*.

April 2022 "*Real-time Characterization of Phase and Twin Variants in Dynamically Deformed Microstructures*" **A. Mishra**, K. Ma, and Avinash M. Dongare, *2022 Mach Conference*.

April 2021 "*Modeling of Multiphase Metallic Materials and Virtual Characterization*" **A. Mishra**, C. Kunka, M. J. Echeverria, R. Dingreville, and Avinash M. Dongare, *2021 CREDDS Center Scientific Advisory Committee (CSAC) meeting*.

Mar 2021 "*Fingerprinting Shock-Induced Deformations via Diffraction*" **A. Mishra**, C. Kunka, M. J. Echeverria, R. Dingreville, and Avinash M. Dongare, *2021 TMS Annual Meeting*.

Feb 2020 "*Slip, Twinning and Phase Transformations in Multiphase Metallic Materials under Shock Loading Conditions*" **A. Mishra**, and Avinash M. Dongare, *2020 TMS Annual Meeting*, San Diego Convention Center in San Diego, California, USA.

Mar 2019 "*Atomistic Origin of Phase Stability in Oxygen Functionalized MXene*" **A. Mishra**, P. Srivastava, A. Carreras, I. Tanaka, H. Mizuseki, K-R. Lee and Abhishek K. Singh, *2019 APS March Meeting*, Boston Convention and Center in Boston, Massachusetts, USA.

Dec 2017 "*Atomistic Origin of Phase Stability in Oxygen Functionalized MXene*" **A. Mishra**, P. Srivastava, A. Carreras, I. Tanaka, H. Mizuseki, K-R. Lee and Abhishek K. Singh, *12^{th} ACCMS VO*, Tohoku University, Sendai, Miyagi, Japan.

Dec 2017 "*Ferroelectricity, Antiferroelectricity, and Ultrathin 2D Electron/Hole Gas in Multifunctional Monolayer MXene*" A. Chandrasekaran, **A. Mishra**, and Abhishek K. Singh, *2017 MRS Fall Meeting*, Hynes Convention Center, Boston, Massachusetts, USA.